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First-principles thermochemistry for the combustion of $TiCl_4$ in a methane flame

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Abstract

This work presents thermochemical data for many of the intermediates in the early stages of the combustion of titanium tetrachloride ($TiCl_4$) in the presence of residual species from a methane flame. An algorithm developed for previous work is employed to ensure that all possible species are considered, reducing the probability of neglecting important species. Thermochemical data and enthalpies of formation are calculated for 141 new species using density functional theory (DFT) and statistical mechanics. Equilibrium calculations are performed to try to identify which intermediates are likely to be most prevalent in the high temperature industrial process. The 10 species with the highest equilibrium compositions are identified and presented in the paper. These species, $TiCl_3OH$, $TiCl_2OH$, $TiCl(OH)_2$, TiOOH, TiOClOH, TiClOH, $TiO(OH)_2$, TiHOOH, $TiCl_2(OH)_2$, and TiClHOH, along with the original $Ti_xO_yCl_z$ species are likely to be the most important species involved in the kinetic model. Data for the remaining species is provided as Supplementary data. The abundance of $Ti_iO_jCl_kH_l$ species is high considering the low amount of hydrogen present in the simulation and they are likely to have a significant impact on the process, something that has not been recognized in previous publications.
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1. Introduction

Titanium dioxide (TiO_2) is widely used as a pigment, as a catalyst support, and as a photocatalyst. The combustion of titanium tetrachloride ($TiCl_4$) to synthesize TiO_2 nanoparticles is a multi-million tonne per year industrial process [1]. In this 'chloride' process, purified titanium tetrachloride is oxidized at high temperatures (1500–2000 K) in a pure oxygen plasma or flame to produce TiO_2 particles [2–4]. The overall stoichiometry of this oxidation process is

$$TiCl_4 + O_2 \rightarrow TiO_{2(\text{nanoparticles})} + Cl_{2(g)}. \quad (1)$$

Although the chloride process is a mature technology, which has been used in industry since 1958, understanding of the gas-phase reactions of $TiCl_4$ in a flame remains incomplete [5]. Recently, West et al. have developed a detailed kinetic model for the chloride process using an oxygen plasma, including thermochemical data for the titanium oxychloride species involved [6–9]. When a pure oxygen plasma is used these species will be sufficient for a kinetic model. However, when a flame is used to superheat the oxygen, the residual combustion

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species (in particular water) are likely to interact chemically with the titanium oxychloride species. It would be instructive to determine to what extent H and OH groups influence the intermediate species formed.

Recently there have been a number of experimental studies of TiO_2 particle formation via combustion in hydrocarbon flames. These include studies using $TiCl_4$ in methane [10] and propane flames [11], as well as the related titanium compound titanium tetraisopropoxide (TTIP) in methane flames [12,13]. This work investigates $TiCl_4$ combustion in a methane flame. This decision was made because of the industrial use of $TiCl_4$ and the well understood nature of a methane flame which includes a detailed kinetic model [14]. Although this work concentrates on $TiCl_4$ in a methane flame the species investigated will be equally useful for the study of $TiCl_4$ and TTIP combustion in other flame environments.

The aim of this study is to provide gas-phase thermochemical data for important titanium species containing oxygen, chlorine and hydrogen ($Ti_iO_jCl_kH_l$). This data has been calculated from first principles using density functional theory (DFT) and statistical mechanics for an exhaustive list of 141 species. In order to ascertain the importance of particular species, equilibrium calculations have been used to identify the most abundant species in a methane flame environment.

2. Computational method

2.1. Species generation

No reaction scheme is available for the combustion of $TiCl_4$ in a hydrocarbon flame. It is therefore necessary to determine those species that could possibly be formed in such a system. This section describes the method we have used to generate an exhaustive list of all likely reacting species. We have recently developed a script to automatically generate species from a selection of existing species [9,15].

Specifically, the method used here involved replacing chlorine atoms on a set of titanium oxychloride ($Ti_xO_yCl_z$) species with either hydrogen atoms or hydroxide groups. When generating the species, the only operation the algorithm performs at any given stage is the replacement of a single-bonded chlorine atom with a single-bonded hydrogen atom or a single bonded hydroxide group. We therefore used the same spin multiplicity as calculated for the original species of West et al. [6,8].

Every possible combination of chlorine replacements in the original $Ti_xO_yCl_z$ species is considered. Figure 1 shows the result of applying this algorithm to the starting species $TiCl_4$. In order to remove any duplicates, an International Chemical Identifier (InChI) name, which can uniquely identify molecules of the same connectivity, is assigned to each species using OpenBabel [16,17]. Duplicate species can then be automatically removed from the set by simply comparing the InChI names.

Each step of this process is fully automated using a combination of Python and Perl scripts. All the new species generated were checked manually to ensure no important species were absent. Conversely, this check also ensured that physically unrealistic species were identified and removed.

2.2. Quantum chemistry calculations

Geometry optimization and molecular properties for all new species were calculated using density functional theory (DFT). Previous studies on the $Ti_xO_yCl_z$ system [6,8,9] indicate that the B97-1 [18] functional is the most reliable and accurate. Consequently, we have used this functional throughout the present study.

The basis set used for all DFT calculations was 6-311+G(d,p). This consists of the 6-311G basis set [19] for oxygen; the McLean–Chandler (12s, 9p) (621111,52111) "negative ion" basis set [20] for chlorine; the Wachters–Hay all-electron basis set [21,22] for titanium, using the scaling factors of Raghavachari and Trucks [23] and supplementary polarization functions and diffuse functions. Such a supplemented, triple-ζ basis set should be large enough to ensure that the basis set truncation error is comparable with, or smaller than, the inherent errors in DFT [24].

All DFT calculations were carried out using the GAUSSIAN03 package [25]. The convergence threshold for the self consistent field was set to 10^{-8} Hartrees. The geometry optimization was considered to have converged if the root-mean-squared force on every atom was under 10^{-5} Hartrees/Bohr.

2.3. Statistical mechanics and equilibrium composition

Heat capacities (C_p°), thermal enthalpy ($H(T)-H(0\ \mathrm{K})$), and entropies (S°) were calculated for temperatures in the range 20–4000 K using the rigid rotator harmonic oscillator (RRHO) approximation, taking unscaled vibrational frequencies and rotational constants from the B97-1/6-311+G(d,p) calculations. The contribution of the excited electronic states to the partition function is ignored as in previous studies [6,8,9], where titanium containing-species have been shown to not possess low energy excited states.

Polynomials in the NASA [26] form were fitted to $C_p^\circ(T)/R$, H° and S° over the temperature ranges $20 - T_x$ K and $T_x - 4000$ K, constrained to ensure that all three functions are continuous and smooth across the boundary temperature, T_x, which was varied to optimize the fit. Using these NASA polynomials, the equilibrium composition as a function of temperature was calculated using the open source software Cantera [27].

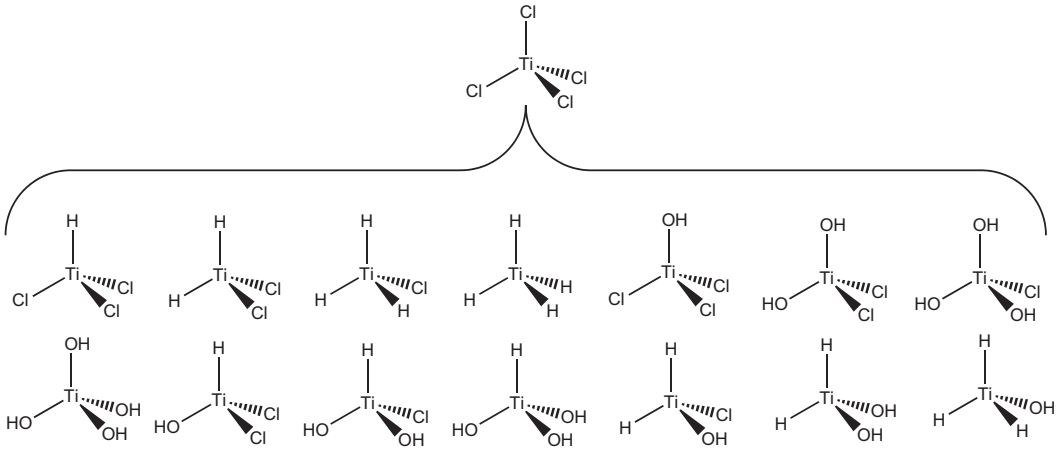


Fig. 1. Generating possible hydrogen-containing species from $TiCl_4$. Thermochemical data for all of these species is provided as Supplementary data.

3. Results and discussion

3.1. Geometries

Converged geometries and molecular properties were found for 141 new species. All this information in the form of GAUSSIAN03 output files and MDL Molfiles is provided in the Supplementary data where species are identified by their unique SMILES names [28,29] generated using OpenBabel [16]. For a small number of DFT calculations the self-consistent field (SCF) did not converge within 400 iterations. Such species were investigated manually and in all such cases were deemed to be physically unreasonable. Typically they involved two atoms within half an Ångstrom of each other or involved one species which had split into two molecules but retained the now unphysical multiplicity. These species accounted for approximately 10% of the total. A challenge for future studies will be to develop a system for automatically rejecting such species. For very large chemical systems it may be sensible to use a cheaper computational method in the first instance to perform this check.

In cases where multiple isomers exist for a given empirical formula, all isomers are included. This is because before running equilibrium composition calculations there is no straightforward way to determine which species are more important. Additionally, since they will take part in different reaction paths, a future kinetic model may need to consider each isomer individually.

3.2. Enthalpy of formation

The standard enthalpy of formation at 0 K, $\Delta_f H^\circ_{0\,K}$, can be determined by subtracting the energy of the species from the energy of its elements in their standard states as calculated using DFT. However, this can lead to systematic errors in the DFT calculations propagating through to the final enthalpies. An alternative method is to use isodesmic reactions [6,15], which relate the species with unknown enthalpy to other species with known enthalpy. In an isodesmic reaction the type and number of bonds broken are the same as those of bonds formed. As the products and reactants have similar electronic structures, systematic errors associated with given bonds are largely canceled out [30].

In this work, we have tried to relate all of the new species using isodesmic reactions with species with known enthalpy from the literature. It has not always been possible to find purely isodesmic reactions as there is not a sufficiently large pool of species with known enthalpy to allow such reactions to be found. In particular, we have not been able to find species with known enthalpy containing the Ti–H bond, which means isodesmic reactions cannot be found for species containing this bond. The pool of species with known enthalpy is given in Table 1. In the case where an isodesmic reaction is not available, an alternative reaction has been found which minimizes the total number of different bonds between products and reactants. Multiple reactions have been found with the lowest number of bond changes and the final enthalpies represent the enthalpy of formation averaged over all the possible reactions. A full list of enthalpies of formation for the 141 new species is given in the Supplementary material, and a subset corresponding to the 10 most abundant species in the equilibrium calculations (Section 3.4) is given in Table 2.

The standard deviations associated with the standard enthalpies of formation for the new species vary widely from 5 kJ/mol to 122 kJ/mol. For the

Table 1
Literature values for standard enthalpies of formation at 298.15 K used to find reactions with low numbers of bond changes to calculate standard enthalpies of formation for new species. Errors are included where available.

Species	$\Delta_f H^\circ_{298.15\ K}$ kJ/mol	Error kJ/mol	Reference
TiOCl	−274	–	[6]
$TiOCl_2$	−598	–	[6]
$TiOC_3$	−639	–	[6]
TiO_2Cl_2	−558	–	[6]
TiO_2Cl_3	−774	–	[6]
$Ti_2O_2Cl_3$	−1257	–	[6]
$Ti_2O_2Cl_4$	−1552	–	[6]
$Ti_2O_3Cl_2$	−1331	–	[6]
$Ti_2O_3Cl_3$	−1418	–	[6]
$Ti_2O_2Cl_5$	−1272	–	[8]
$Ti_2O_2Cl_6$	−1503	–	[8]
$TiCl_4$	−763.16	–	[32]
$TiCl_3$	−539.32	–	[32]
$TiCl_2$	−237.23	–	[32]
TiO_2	−305.43	–	[32]
ClOCl	87.86	–	[32]
OClO	104.60	–	[32]
ClOH	−74.48	–	[32]
H_2O	−241.826	±0.040	[33]
HCl	−92.31	±0.10	[33]
OH	38.99	–	[32]
Cl	121.301	±0.008	[33]
O	249.18	±0.10	[33]

majority of species with a large standard deviation it is mainly due to one reaction giving a markedly different enthalpy to all the others. This is due to one or more of the known species having a spurious enthalpy. This is not unexpected as inconsistences in experimentally obtained enthalpies of formation have previously been observed [6].

At high temperature, where entropy is the dominant contribution to the Gibbs energy, the equilibrium compositions are relatively insensitive to the enthalpy of formation. While standard deviations of up to 100 kJ/mol in the enthalpy are significant, the large errors tended to be associated with unstable species (Section 3.4). These values correspond to a standard deviation in the mean of up to 40 kJ/mol. Errors of this scale are expected and are broadly consistent with those obtained by Phadungsukanan et al. [15] for silica species. A basic sensitivity analysis was conducted whereby enthalpies of these species were varied by this amount and, while the equilibrium concentrations were affected, the conclusions remained unchanged.

3.3. Thermochemistry

A full list of computed molecular and standard thermochemical data for species convergent under geometry optimization using the B97-1 functional with the 6-311+G(d,p) basis set are presented in the Supplementary material. Table 3 shows this data for the 10 most abundant species in the equilibrium composition (Section 3.4).

3.4. Equilibrium composition

Equilibrium compositions were calculated using NASA polynomials for: 141 new species as calculated in this work, those from the titanium oxychloride reaction mechanism presented by West et al. [8], and all of the species except those containing nitrogen from the GRI-Mech methane mechanism [14]. The West et al. reaction mechanism includes NASA polynomials for $TiCl_x$ ($x = 1, \ldots, 4$), TiO, TiO_2, chlorine oxides and standard gases, taken from the NASA database [31]. Additionally the polynomials for HCl from the NASA database were used. The GRI-Mech species were used because they include all important residual combustion species. For species which exist in both the GRI-Mech mechanism and the West et al. mechanism, the data used is taken from West et al.

Table 2
Standard enthalpies of formation at 0 K and 298.15 K, with standard deviation and number of reactions used with the corresponding number of bond changes for the 10 highest-concentration species calculated at the B97-1/6-311+G(d,p) level of theory for stable electronic ground states. To find corresponding information in the Supplementary material use the SMILES names provided in Fig. 3.

Species	$\Delta_f H^\circ_{0\ K}$ kJ/mol	$\Delta_f H^\circ_{298.15\ K}$ kJ/mol	Standard deviation kJ/mol	No. of reactions	No. of bond changes
$TiCl_3OH$	−941	−917	14	6	0
$Ti_2O_2Cl_3OH$	−1741	−1708	14	4	0
$Ti_2OCl_4(OH)_2$	−1936	−1893	56	6	0
$TiCl_2OH$	−719	−699	24	6	0
TiOClOH	−742	−724	34	7	0
$TiCl_2(OH)_2$	−1079	−1054	28	6	0
$TiOCl_2OH$	−793	−770	28	6	0
Ti_2O_3ClOH	−1545	−1518	64	6	0
$Ti_2O_2Cl_2(OH)_2$	−1866	−1831	43	6	0
$TiCl(OH)_2$	−850	−829	29	6	0

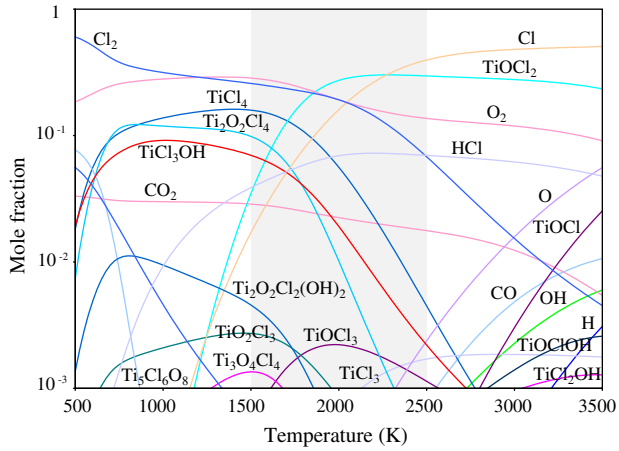
Table 3
Standard entropy at 298.15 K, $S^{\circ}_{298.15\ K}$, and molecular properties for the 10 highest-concentration species calculated at the B97-1/6-311+G(d,p) level of theory for stable electronic ground states. The electronic degeneracy, g, and the rotational symmetry, s, are included. To find corresponding information in the Supplementary material use the SMILES names provided in Fig. 3.

Species	$S^{\circ}_{298.15\ K}$ J/mol K	g	s	Rot. const. GHz			Vibrational frequencies cm^{-1}
$TiCl_3OH$	386.0	1	1	1.5727	1.5474	1.1256	101.06, 113.62, 115.97, 136.66, 157.85, 200.48, 224.21, 402.83, 483.16, 492.26, 828.6, 3944.84
$Ti_2O_2Cl_3OH$	460.4	1	1	1.1879	0.5062	0.3801	43.34, 72.96, 97.9, 109.55, 109.62, 157.42, 165.47, 219.24, 245.55, 278.26, 321.47, 330.37, 432.67, 441.93, 508.43, 545.41, 716.4, 735.29, 779.19, 809.18, 3941.09
$Ti_2OCl_4(OH)_2$	575.2	1	1	0.7414	0.2997	0.2867	5.01, 18.38, 21.39, 84.21, 106.09, 109.98, 116.43, 126.67, 137.77, 151.28, 161.67, 190.33, 232.29, 249.37, 274.81, 284.7, 342.19, 407.01, 439.06, 485.41, 488.62, 498.47, 773.14, 812.75, 899.08, 3924.35, 3948.95
$TiCl_2OH$	351.3	2	2	3.9435	1.9430	1.3017	89.61, 101.74, 134.39, 166.55, 367.6, 383.37, 478.74, 794.8, 3973.44
TiOClOH	331.6	1	1	7.5435	2.6268	1.9484	34.4, 145.65, 224.39, 420.53, 519.71, 548.94, 751.0, 1079.95, 3919.73
$TiCl_2(OH)_2$	383.5	1	1	2.4046	1.6691	1.3998	108.56, 120.51, 153.55, 156.39, 183.08, 188.95, 263.3, 293.99, 375.98, 434.04, 480.5, 777.73, 812.37, 3925.11, 3943.23
$TiOCl_2OH$	394.7	2	1	2.4723	1.7180	1.3842	21.26, 120.99, 126.31, 131.29, 166.06, 181.5, 345.92, 424.07, 479.52, 678.21, 804.42, 3945.81
Ti_2O_3ClOH	409.3	1	1	2.2879	0.8596	0.7099	67.67, 117.03, 140.1, 151.2, 162.95, 163.59, 225.91, 266.65, 313.61, 386.18, 468.56, 496.52, 699.72, 729.63, 755.33, 777.45, 1060.05, 3986.56
$Ti_2O_2Cl_2(OH)_2$	467.8	1	1	1.4469	0.5871	0.4531	49.32, 83.22, 112.76, 116.13, 119.33, 140.78, 164.85, 193.31, 223.08, 240.26, 260.1, 261.49, 323.14, 366.32, 440.59, 447.6, 545.44, 707.53, 739.88, 764.57, 772.79, 812.27, 3963.01, 3966.45
$TiCl(OH)_2$	360.3	2	1	5.6454	2.7264	1.8385	88.14, 100.44, 134.4, 160.77, 272.28, 352.64, 386.3, 413.72, 737.37, 784.81, 3989.12, 3993.19

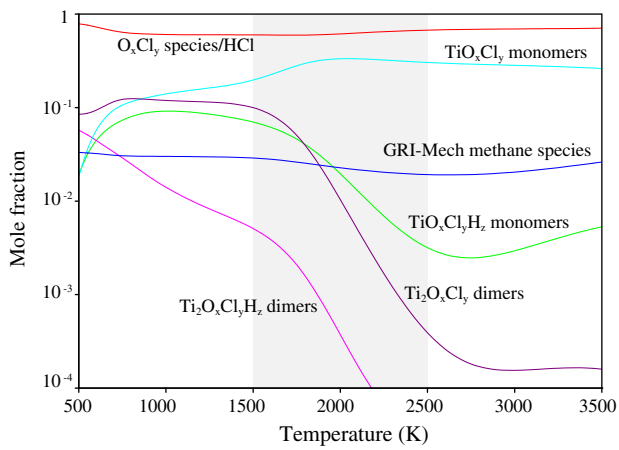
Computed equilibrium data calculated over the temperature range of 500–3500 K are shown in Fig. 2 for a fuel lean mixture initially containing 48.5 mol% $TiCl_4$, 3 mol% CH_4, and 48.5 mol% O_2 at 101.3 kPa. These conditions are typical in industry where a lean hydrocarbon flame is used to superheat the oxygen and the reaction with $TiCl_4$ is close to stoichiometric. Figure 2(a) shows only the upper part of the concentration plot in order to show the specific species that are most stable. This plot clearly shows that species containing both titanium and hydrogen are of considerable importance. This is true for all of the flame environments investigated. For clarity, the individual species concentrations have been collected in Fig. 2(b). The following groups were used: GRI-Mech species (excluding O, O_2, O_3, taken from West et al.), West et al. species split into TiO_xCl_y monomers, $Ti_2O_xCl_y$ dimers, O_xCl_y species/HCl, and the new species presented in this work separated into $TiO_xH_yCl_z$ monomers and $Ti_2O_xH_yCl_z$ dimers. The precise species used in each group are tabulated in the Supplementary material which includes references to the original sources. There are 201 species overall in the system, some 141 of which are those presented in this work.

Geometries of the 10 most abundant species containing hydrogen (in terms of mole fraction) over the temperature range 1500–2500 K are shown in Fig. 3. Both empirical formula and SMILES format names are given to help the reader associate empirical names used in the paper with the SMILES used in the Supplementary material. A number of these hydrogen containing species are present at comparable or greater levels than the titanium oxychloride species. In an industrial context, where the gases are essentially at equilibrium because particle formation is rate-limiting, some of these species are likely to play an important role in the gas kinetics and nucleation process.

A number of different initial compositions were experimented with, including methane lean flames



(a) Detailed equilibrium concentrations



(b) Collated equilibrium concentrations

Fig. 2. Computed equilibrium for combustion of $TiCl_4$ in a methane flame. Initial concentrations are 48.5 mol% $TiCl_4$, 3 mol% CH_4, and 48.5 mol% O_2 at 101.3 kPa. Thermochemical data for $Ti_xO_yCl_z$ species from West et al. [8] and the NASA database [31]. All species from the GRI-Mech methane mechanism included [14]. The central grey band indicates the temperature range most relevant to industrial reactors.

with very low $TiCl_4$ concentrations, as typically used in experiments, methane rich flames with larger $TiCl_4$ concentrations, as well as methane lean flames with larger $TiCl_4$ concentrations ($Ti:O_2 = 1:1$) which have industrial relevance. As expected the relative abundance of hydrogen in the initial composition has a large impact on the species present at equilibrium. The more hydrogen present, the higher the concentrations of the new $Ti_iO_jCl_kH_l$ species compared to the existing $Ti_iO_jCl_k$ species. Conversely, the higher the initial concentration of oxygen, the lower the concentration of the new species and additionally within the new species hydroxide groups are preferred to the hydrides. Specifically, at a typical industrial reactor temperature of 2000 K the new monomers and dimers become more abundant than their titanium oxychloride counterparts when the initial CH_4 concentration rises above 3.8% compared to stoichiometric $TiCl_4:O_2$ conditions. For this reason we recommend that the original kinetic model presented by West et al. is only applicable when hydrogen-containing species are at very low concentrations.

For the initial conditions presented in Fig. 2, where the amount of hydrogen in the system is very low, the high concentrations of some hydrogen-containing species indicate the comparable thermodynamic stability of Ti–OH bonds with Ti–Cl bonds. $TiCl_3OH$ is the most abundant new species and is comparable to $TiCl_4$ and HCl and will be important to the system reaction kinetics. Of the 10 most abundant new species, four are titanium dimers and therefore the presence of

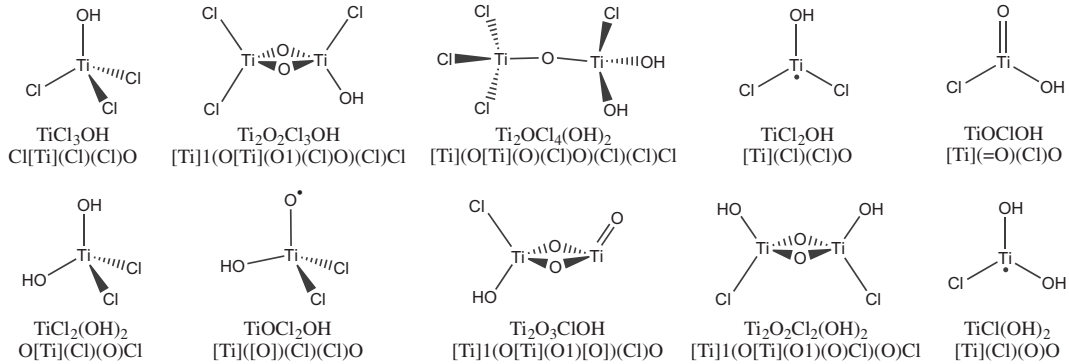


Fig. 3. Geometries of the 10 most abundant new species over the temperature range 1500–2500 K. The Supplementary data is all organized in terms of the SMILES system for naming molecules. These are reported here so that the reader can associate the more readable names used in the paper with the SMILES used in the Supplementary data.

hydrogen is highly likely to play an important role in dimerization and subsequent particle growth.

4. Conclusions

This work has extended the detailed thermochemistry of the gas-phase titanium oxychloride species investigated by West et al. [6] to a similar system of hydrogen-containing species. These were not previously included in their work but play an important role in the industrial flame synthesis of TiO_2 where small amounts of residual combustion species are present. New molecular geometries have been proposed through automated species generation and investigated using density functional theory with the B97-1 functional and the 6-311+G(d,p) basis set. This enabled detailed thermochemical data to be computed for the generated species, which have not been reported in the literature and are difficult to obtain experimentally.

Using the thermochemical data, temperature-dependent equilibrium compositions were obtained through simulations at industrial operating conditions. A number of conclusions are made from these studies:

- Hydrogen containing species are stable and will exist at industrial and experimental conditions.
- $TiCl_3OH$, $TiCl_2OH$, $TiCl(OH)_2$, TiOOH, TiOClOH, TiClOH, $TiO(OH)_2$, TiHOOH, $TiCl_2(OH)_2$, and TiClHOH are highlighted as 10 of the most stable $Ti_iO_jCl_kH_l$ species for industrial conditions.
- Specifically $TiCl_3OH$ is the most abundant new species. It is hence likely to play an important role in the combustion synthesis of TiO_2 nanoparticles even when there is a relatively small amount of hydrogen, as in industrial flame reactors.

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Appendix A. Supplementary data

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.011.

References

[1] J. Emsley, *Molecules at an Exhibition*, Oxford University Press, 1999.
[2] R.A. Gonzalez, C.D. Musick, J.N. Tilton, *Process for controlling agglomeration in the manufacture of TiO_2*, US Patent 5,508,015, 1996.
[3] J.C. Deberry, M. Robinson, M. Pomponi, A.J. Beach, Y. Xiong, K. Akhtar, *Controlled vapor phase oxidation of titanium tetrachloride to manufacture titanium dioxide*, US Patent 6,387,347, 2002.
[4] J.C. Magyar, A.J. Morris, G.D. Wootten, W.A. Yuill, *Method and apparatus for producing titanium dioxide*, US Patent 6,207,131, 2006.
[5] B. Karlemo, P. Koukkari, J. Paloniemi, *Formation of gaseous intermediates in titanium(IV) chloride plasma oxidation*, *Plasma Chem. Plasma Process.* 16 (1996) 59–77.
[6] R.H. West, G.J.O. Beran, W.H. Green, M. Kraft, *First-principles thermochemistry for the production of* TiO_2 *from* $TiCl_4$, *J. Phys. Chem. A* 111 (18) (2007) 3560–3565.
[7] R.H. West, M.S. Celnik, O.R. Inderwildi, M. Kraft, G.J.O. Beran, W.H. Green, *Toward a comprehensive model of the synthesis of* TiO_2 *particles from* $TiCl_4$, *Ind. Eng. Chem. Res.* 46 (19) (2007) 6147–6156.
[8] R.H. West, R.A. Shirley, M. Kraft, C.F. Goldsmith, W.H. Green, *A detailed kinetic model for*

combustion synthesis of titania from $TiCl_4$, *Combust. Flame* 156 (2009) 1764–1770.

[9] R. Shirley, Y. Liu, T.S. Totton, R.H. West, M. Kraft, *First-principles thermochemistry for the combustion of a* $TiCl_4$ *and* $AlCl_3$ *mixture*, *J. Phys. Chem. A* 113 (49) (2009) 13790–13796.

[10] M. Katzer, A. Weber, G. Kasper, *The effects of electrical fields on growth of titania particles formed in a* CH_4–O_2 *diffusion flame*, *J. Aerosol Sci.* 32 (9) (2001) 1045–1067.

[11] H. Xie, G. Gao, Z. Tian, N. Bing, L. Wang, *Synthesis of* TiO_2 *nanoparticles by propane/air turbulent flame CVD process*, *Particuology* 7 (3) (2009) 204–210.

[12] O. Arabi-Katbi, S. Pratsinis, P. Morrison, C. Megaridis, *Monitoring the flame synthesis of TiO2 particles by in-situ FTIR spectroscopy and thermophoretic sampling*, *Combustion Flame* 124 (4) (2001) 560–572.

[13] H. Kammler, R. Jossen, P. Morrison, S. Pratsinis, G. Beaucage, *The effect of external electric fields during flame synthesis of titania*, *Powder Technol.* 135 (1) (2003) 310–320.

[14] G.P. Smith, D.M. Golden, M. Frenklach, et al., GRI-Mech 3.0, Software Package, 2008, available at http://me.berkeley.edu/gri-mech/.

[15] W. Phadungsukanan, S. Shekar, R. Shirley, M. Sander, R.H. West, M. Kraft, *First-principles thermochemistry for silicon species in the decomposition of tetraethoxysilane*, *J. Phys. Chem. A* 113 (31) (2009) 9041–9049.

[16] R. Guha, M.T. Howard, G.R. Hutchison, et al., *The blue obelisk interoperability in chemical informatics*, *J. Chem. Inf. Model.* 46 (3) (2006) 991–998.

[17] N. O'Boyle, C. Morley, G. Hutchison, *Pybel: a python wrapper for the openbabel cheminformatics toolkit*, *Chem. Cent. J.* 2 (1) (2008) 5.

[18] F.A. Hamprecht, A.J. Cohen, D.J. Tozer, N.C. Handy, *Development and assessment of new exchange-correlation functionals*, *J. Chem. Phys.* 109 (15) (1998) 6264–6271.

[19] R. Krishnan, J.S. Binkley, R. Seeger, J.A. Pople, *Self-consistent molecular-orbital methods. xx. Basis set for correlated wave-functions*, *J. Chem. Phys.* 72 (1980) 650–654.

[20] A.D. McLean, G.S. Chandler, *Contracted gaussian-basis sets for molecular calculations. i. 2nd row atoms,* $z = 11$–18, *J. Chem. Phys.* 72 (1980) 5639–5648.

[21] A.J. Wachters, *Gaussian basis set for molecular wavefunctions containing third-row atoms*, *J. Chem. Phys.* 52 (1970) 1033–1036.

[22] P.J. Hay, *Gaussian basis sets for molecular calculations. the representation of 3D orbitals in transition-metal atoms*, *J. Chem. Phys.* 66 (1977) 4377–4384.

[23] K. Raghavachari, G.W. Trucks, *Highly correlated systems – excitation-energies of 1st row transition-metals Sc–Cu*, *J. Chem. Phys.* 91 (1989) 1062–1065.

[24] A.D. Boese, J.M.L. Martin, N.C. Handy, *The role of the basis set: assessing density functional theory*, *J. Chem. Phys.* 119 (6) (2003) 3005–3014.

[25] M.J. Frisch, G.W. Trucks, H.B. Schlegel, et al., Gaussian 03, Revision C.02, 2004, Gaussian, Inc., Wallingford, CT, 2003.

[26] S. Gordon, B.J. McBride, Computer program for calculation of complex chemical equilibrium composition, rocket performance, incident and reflected shocks and chapman-jouguet detonations, Tech. Rep. NASA-SP-273, NASA, Glenn Research Center, Cleveland, OH, 1976, available at http://hdl.handle.net/2060/19780009781.

[27] D.G. Goodwin, An open source, extensible software suite for CVD process simulation, in: M. Allendorff, F. Maury, F. Teyssandier (Eds.), *Chemical Vapor Deposition XVI and EUROCVD 14, Vol. 2003–08 of ECS Proceedings*, The Electrochemical Society, Pennington, New Jersey, USA, 2003, pp. 155–162, available at code.google.com/p/cantera/.

[28] D. Weininger, *SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules*, *J. Chem. Inf. and Comput. Sci.* 28 (1) (1988) 31–36.

[29] D. Weininger, A. Weininger, J. Weininger, *SMILES. 2. Algorithm for generation of unique SMILES notation*, *J. Chem. Inf. Comput. Sci.* 29 (2) (1989) 97–101.

[30] W.J. Hehre, R. Ditchfield, L. Radom, J.A. Pople, *Molecular orbital theory of the electronic structure of organic compounds. v. molecular theory of bond separation*, *J. Am. Chem. Soc.* 92 (16) (1970) 4796–4801.

[31] B.J. McBride, M.J. Zehe, S. Gordon, Tech. Rep. TP-2002-211556, NASA, Glenn Research Center, Cleveland, OH, 2002, available at http://hdl.handle.net/2060/20020085330.

[32] M.W. Chase Jr., NIST-JANAF thermochemical tables, fourth edition, J. Phys. Chem. Ref. Data Monograph 9 (1998) I–II.

[33] J. Cox, D. Wagman, V. Medvedev, *CODATA Key Values for Thermodynamics*, Hemisphere Publishing Corp., New York, 1984.